SUPPLEMENTARY REPORT

Two-Agent Generation and Verification of Negative Differential-Privacy Questions

Anonymous artifact report

Abstract

This report documents one completed run of an isolated two-agent workflow for constructing false, single-mutation differential-privacy questions from accepted positive questions. Agent A generated one proof-guided mutation per source using only positive inputs and cited proof context. Agent B then independently assessed truth, semantic locality, proof validity, and novelty without receiving Agent A's expected label, expected failure, confidence, or acceptance signal. Of ten generated candidates, all ten were verified false and single-component; nine satisfied the novelty gate and were accepted, while one mathematically false duplicate was rejected. No case entered the formal human-review queue. The report states the information boundary, decision rule, run-level evidence, and representative mathematical checks.

1 Scope and Experimental Unit

The protected input is `accepted_positive_questions.json`, containing source IDs $1, \ldots, 10$. The workflow requires exactly one candidate per source and prohibits silent omission, extra sources, and placeholders. The source was treated as immutable; its SHA-256 digest was

`18a9f309bdda960c7a2c3fae5932b8f14c807ff85df27567949664ca3ee53845`.

No external GPT API was used by the workflow. The unit of generation was an exact textual replacement of one declared span in one semantic component: an assumption, an algorithm step, or a conclusion.

Table 1: Observed outcome of the completed run. Candidate coverage is distinct from accepted-set coverage: source 9 produced a valid false claim but was excluded as a duplicate.

Measure	Value	Measure	Value
Target positive sources	10	Generated candidates	10
Candidate source coverage	10/10	Agent B results	10/10
Verified false, single-change	10/10	Accepted negatives	9
Rejected as duplicate	1	Human-review cases	0
Generated mutation mix	3/4/3	Accepted mutation mix	3/3/3

Mutation order is assumption / algorithm / conclusion.

2 Protocol and Information Boundary

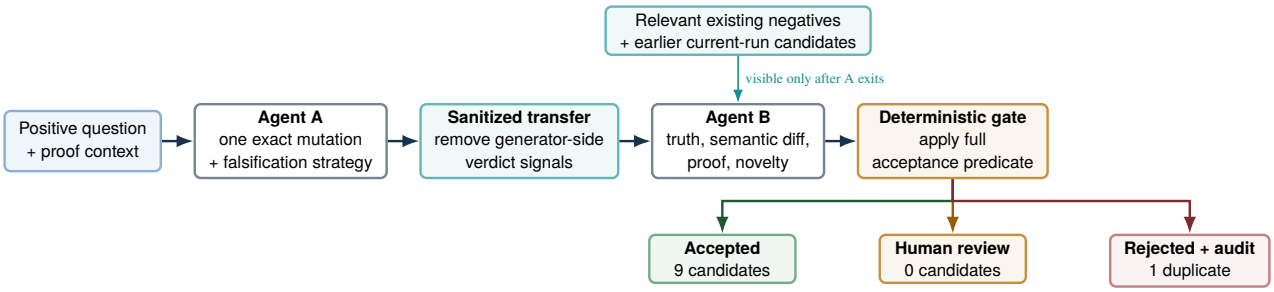


Figure 1: Information flow. Existing negatives enter only Agent B's verification packet. Agent A does not assign a final truth, novelty, verification, or acceptance label.

2.1 Agent A: positive-only generation

Agent A ran in a filesystem sandbox whose readable benchmark workspace was restricted to the specification, `input/positive_sources.json`, and `input/source_context.json`. The sandbox excluded the protected source path, label-0 examples, mutation history, previous audits and reports, Agent B files, and the surrounding repository. An isolation probe recorded successful reads of all three allowed inputs and denied reads or listings for the protected source, history, and repository.

For each source, Agent A identified a privacy-critical proof dependency, replaced one exact `before` span with one `after` span, preserved unrelated semantics, and proposed a concrete falsification route. Its output contained no final truth, novelty, verification, or acceptance field. All ten `preliminary_novelty_matches` arrays were empty.

2.2 Agent B: independent adjudication

Agent B received the original question, cited proof context, candidate question, exact mutation, proposed strategy, deterministic similarity data, relevant existing negatives, and earlier current-run candidates. The transfer excluded Agent A's expected failure, expected label, confidence, acceptance signals, hidden reasoning, and historical verifier decisions.

Agent B independently returned a truth assessment, the single-component check, a novelty status, a full proof, retained-assumption checks, uncertainty fields, and a decision. Falsity required an explicit counterexample, an exact privacy-loss or divergence calculation, an applicable tight bound, or a formal reduction to a concrete privacy violation. Merely observing that the original proof no longer applied was insufficient.

2.3 Acceptance and routing

For a candidate c, the deterministic merger accepted exactly when

$$\begin{aligned}\mathcal{A}(c) = 1 \iff\ & [\text{claim}(c) = \text{FALSE}] \wedge [\text{single}(c) = \text{true}] \\ & \wedge [\text{novelty}(c) \in \{\text{NOVEL}, \text{NOVEL_BUT_RELATED}\}] \\ & \wedge [\text{verification}(c) = \text{VERIFIED}] \wedge [\text{decision}(c) = \text{ACCEPT}] \wedge [\text{uncertainties}(c) = \varnothing]. \end{aligned} \tag{1}$$

Ambiguous, incomplete, delicate, multi-component, or disputed cases were designated for human review. A verified false duplicate was rejected, not routed to review.

3 Run-Level Evidence

Agent A generated three assumption, four algorithm, and three conclusion perturbations. Agent B marked all ten claims FALSE, VERIFIED, single-component, and uncertainty-free. Novelty adjudication yielded four NOVEL, five NOVEL BUT RELATED, and one DUPLICATE. The duplicate was the only rejection, leaving an exactly balanced accepted set of three mutations per component type. Across the ten verifier records, the audit stores 40 explicit retained-assumption checks.

Table 2: Per-candidate decision trace. Every row was assessed false, verified, single-component, and uncertainty-free. "Related" abbreviates `NOVEL_BUT_RELATED`.

Source	Mutation	Novelty	Proof mode	Decision
1	Assumption	Related	Explicit hockey-stick counterexample	Accept
2	Conclusion	Related	Exact privacy-loss calculation	Accept
3	Algorithm	Novel	Exact variance calculation	Accept
4	Algorithm	Related	Exact Gaussian trade-off counterexample	Accept
5	Assumption	Novel	Explicit trade-off counterexample	Accept
6	Conclusion	Novel	Exact Gaussian trade-off calculation	Accept
7	Conclusion	Related	Explicit Gaussian-DP counterexample	Accept
8	Assumption	Novel	Explicit sensitivity counterexample	Accept
9	Algorithm	Duplicate	Exact privacy-loss calculation	Reject
10	Algorithm	Related	Exact moment calculation	Accept

The proof modes comprise six exact calculations and four explicit counterexamples. Relatedness was not treated as duplication: sources 1, 2, and 7 were related to visible existing negatives, while sources 4 and 10 were related to an earlier current-run candidate. Source 9 matched `generated:fresh_uncovered_run:5` and failed only the novelty clause

in Eq. (1).

4 Representative Mathematical Verifications

The following cases span all three mutation types. Each proof fixes concrete parameters, checks the retained domains, and evaluates the changed claim directly.

4.1 Assumption mutation: independence is privacy-critical

Candidate `newneg_source_1_assumption_1` VERIFIED

Before	$Z_1, \dots, Z_m$ are i.i.d. real noises.
After	$Z_1, \dots, Z_m$ are identically distributed real noises.

Take $m = 2$, neighboring data sets with score vectors

$$(f_1(X), f_2(X)) = (1, 0), \qquad (f_1(X'), f_2(X')) = (0, 1),$$

and perfectly dependent noises $Z_1 = Z_2 = Z$, where $Z \sim \mathcal{N}(0, 100)$. The marginal density and both sensitivity bounds are unchanged, but

$$M(X) = 1, \qquad M(X') = 2 \quad \text{almost surely.}$$

Thus $H_1(M(X) \| M(X')) = 1$. At $t = 0$, the largest hockey-stick divergence between Gaussian shifts separated by at most two is

$$\delta_2(0) = 2\Phi(0.1) - 1.$$

Since

$$\Phi(0.1) - \tfrac{1}{2} = \int_0^{0.1} \phi(u)\, du < \frac{0.1}{\sqrt{2\pi}} < 0.04,$$

the proposed upper bound satisfies $m\delta_2(0) < 0.16 < 1$. The mutation is therefore false, and the counterexample isolates independence as the only removed dependency.

4.2 Algorithm mutation: changing the released count

Candidate `newneg_source_4_algorithm_1` VERIFIED

Before	$A(X) = \big(f(X) + \eta \mathbf{1}_d + Z,\ n + 2\eta\big)$.
After	$A(X) = \big(f(X) + \eta \mathbf{1}_d + Z,\ n + \eta\big)$.

Set $d = 1$ and add the valid row $x = 0$. The neighboring output means differ by $v = (0, 1)^\top$. With all stated Gaussian laws and independence retained, the common covariance is

$$\Sigma = \mu^{-2} \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}, \qquad \Sigma^{-1} = \mu^2 \begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix}.$$

Therefore

$$v^\top \Sigma^{-1} v = 4\mu^2,$$

so the neighboring Gaussian experiment has trade-off $G_{2\mu}$, not a function bounded below by G_μ. At type-I error $\alpha = 1/2$,

$$G_{2\mu}(1/2) = \Phi(-2\mu) < \Phi(-\mu) = G_\mu(1/2) \qquad (\mu > 0).$$

This pointwise violation disproves the mutated μ-GDP claim while preserving adjacency, row bounds, and all noise parameters.

4.3 Conclusion mutation: Gaussian composition is Euclidean

Candidate `newneg_source_6_conclusion_1` VERIFIED

Before $G_{\mu_1} \otimes G_{\mu_2} = G_{\sqrt{\mu_1^2+\mu_2^2}}$.

After $G_{\mu_1} \otimes G_{\mu_2} = G_{\mu_1+\mu_2}$.

Choose $\mu_1 = \mu_2 = 1$. The product experiment is

$$\mathcal{N}(0, I_2) \quad \text{versus} \quad \mathcal{N}\big((1,1)^\top, I_2\big).$$

An orthogonal rotation maps the mean displacement to $(\sqrt{2}, 0)^\top$, preserves covariance and testing trade-off, and leaves the second coordinate uninformative. Hence the left side is exactly $G_{\sqrt{2}}$. The mutated right side is G_2, and at $\alpha = 1/2$,

$$G_{\sqrt{2}}(1/2) = \Phi(-\sqrt{2}) > \Phi(-2) = G_2(1/2).$$

The asserted identity is false. This verification identifies the exact experiment rather than relying on failure of the original derivation.

4.4 Negative control: false but not novel

Source 9 changed the K-norm kernel from

$$\exp\left(-\frac{\|y - T(X)\|}{\sigma}\right) \quad \text{to} \quad \exp\left(-\frac{2\|y - T(X)\|}{\sigma}\right).$$

In one dimension this is Laplace noise with scale $\sigma/2$. For a sensitivity-attaining neighboring shift of Δ, the event $(-\infty, 0]$ yields an exact likelihood ratio

$$\frac{\Pr(M(X) \le 0)}{\Pr(M(X') \le 0)} = \exp\left(\frac{2\Delta}{\sigma}\right) > \exp\left(\frac{\Delta}{\sigma}\right),$$

so the retained (Δ/σ)-DP conclusion is false. Agent B nevertheless rejected the candidate because the same doubled-exponent construction already appeared in a visible negative. This case demonstrates that mathematical falsity is necessary but not sufficient for acceptance.

5 Auditability and Reproducibility

The audit binds source, generator, verifier, and merged outputs with SHA-256 digests. It embeds the exact generator candidate, Agent B record, gate checks, human-review triggers, and final disposition for every source. The validator requires exact source coverage and ordering, one declared replacement per candidate, full acceptance-rule compliance, routing of disputed cases, an unchanged protected-source digest, and byte-identical deterministic merges.

Table 3: Hashes of the six delivered workflow artifacts. Paths are relative to `neg_generation/`.

Artifact	SHA-256
`generated_negative_candidates.json`	`a2301f43252dc81b5f25a3c5f5944d4451171a9aa7805615641465bc75c4670e`
`generated_negative_audit.json`	`1c1ac498a681c4e40b83fb55c44f9d5fc3feafbfcbcc06dee7991bba5740fa67`
`verifier_results.json`	`dd3ac9b0ca64621a6d33568fea04e01ee3922ebe94c972f6f7dcf54ce10619c5`
`generation_report.md`	`1bcc7141b9d858e599db8f70363669f72d25574943444f949b4b2f9634b9395b`
`generated_negative_human_review.json`	`37517e5f3dc66819f61f5a7bb8ace1921282415f10551d2defa5c3eb0985b570`
`TWO_AGENT_NEGATIVE_GENERATION_SPEC.md`	`b616dbcf7788e090a67149ebd09593865a69355a7d265f9b43d53f23d5baf83e`